

## HAMAMATSU, Japan

WMO: 476540

Lat: 34.7539N

Lon: 137.7117E

Elev: 48

StdP: 100.75

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 0.1        | 1.0 | -11.3                         | 1.4 | 4.5  | -9.7 | 1.6 | 4.7  | 10.0                  | 6.7  | 9.3 | 6.6  | 4.2                   | 290  | 0.403 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 6.4                    | 33.1            | 25.6 | 31.7 | 25.4 | 30.7 | 25.2 | 26.9                | 30.6 | 26.4 | 29.8 | 26.0 | 29.1 | 5.0                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 26.0                            | 21.5 | 28.7 | 25.6 | 20.9 | 28.4 | 25.1 | 20.4 | 28.0 | 85.0          | 30.8 | 82.9 | 29.7 | 81.0 | 29.2 | 28.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 8.9               | 8.0  | 7.2 | -1.8                       | 36.0 | 1.1                | 1.4 | -2.6                                               | 37.0 | -3.3       | 37.8 | -3.9       | 38.6 | -4.8       | 39.6 |    |    |
|                   |      |     | -4.2                       | 27.7 | 1.0                | 0.7 | -5.0                                               | 28.2 | -5.6       | 28.6 | -6.2       | 29.0 | -6.9       | 29.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 17.0   | 6.3   | 7.1   | 10.2  | 15.0  | 19.4  | 22.6  | 26.6  | 28.0  | 24.9  | 19.8  | 14.1  | 8.8  |
|                                                                                  | DBStd       | 7.80   | 2.31  | 2.72  | 2.88  | 2.81  | 2.03  | 2.07  | 2.25  | 1.77  | 2.43  | 2.56  | 2.85  | 2.85 |
|                                                                                  | HDD10.0     | 299    | 117   | 88    | 33    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 57   |
|                                                                                  | HDD18.3     | 1491   | 373   | 314   | 252   | 103   | 12    | 1     | 0     | 0     | 0     | 13    | 128   | 294  |
|                                                                                  | CDD10.0     | 2837   | 2     | 7     | 40    | 152   | 290   | 377   | 514   | 557   | 447   | 304   | 126   | 21   |
|                                                                                  | CDD18.3     | 989    | 0     | 0     | 0     | 4     | 44    | 128   | 256   | 298   | 197   | 59    | 2     | 0    |
|                                                                                  | CDH23.3     | 7818   | 0     | 0     | 0     | 8     | 125   | 526   | 2314  | 3191  | 1462  | 190   | 2     | 0    |
|                                                                                  | CDH26.7     | 2343   | 0     | 0     | 0     | 0     | 9     | 84    | 724   | 1143  | 363   | 21    | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.4    | 4.2   | 4.1   | 4.0   | 3.7   | 3.2   | 2.9   | 2.8   | 2.9   | 2.9   | 3.2   | 3.9   |      |
| Precipitation                                                                    | PrecAvg     | 1864   | 66    | 89    | 145   | 176   | 196   | 221   | 199   | 140   | 256   | 187   | 124   | 66   |
|                                                                                  | PrecMax     | 2542   | 159   | 226   | 246   | 343   | 306   | 366   | 424   | 413   | 646   | 689   | 235   | 143  |
|                                                                                  | PrecMin     | 1314   | 1     | 23    | 62    | 29    | 85    | 52    | 23    | 23    | 77    | 28    | 15    | 5    |
|                                                                                  | PrecStd     | 333    | 40    | 50    | 44    | 75    | 66    | 69    | 106   | 83    | 131   | 120   | 61    | 41   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 14.9  | 17.6  | 20.6  | 24.5  | 27.9  | 31.0  | 35.1  | 35.8  | 32.8  | 28.8  | 23.2  | 18.7 |
|                                                                                  |             | MCWB   | 10.5  | 13.0  | 13.3  | 16.4  | 19.2  | 23.4  | 25.4  | 26.0  | 24.9  | 21.9  | 17.4  | 14.6 |
|                                                                                  | 2%          | DB     | 13.0  | 15.3  | 18.5  | 22.6  | 26.0  | 28.9  | 32.8  | 33.6  | 31.1  | 26.8  | 21.5  | 16.5 |
|                                                                                  |             | MCWB   | 8.4   | 10.6  | 12.3  | 15.6  | 18.3  | 22.4  | 25.6  | 25.8  | 24.4  | 20.5  | 16.1  | 12.2 |
|                                                                                  | 5%          | DB     | 11.7  | 13.4  | 16.9  | 21.2  | 24.7  | 27.3  | 31.3  | 32.2  | 30.0  | 25.3  | 20.2  | 15.0 |
|                                                                                  |             | MCWB   | 7.1   | 8.6   | 12.1  | 15.1  | 17.9  | 21.9  | 25.4  | 25.8  | 24.3  | 19.9  | 15.5  | 10.4 |
|                                                                                  | 10%         | DB     | 10.5  | 12.0  | 15.5  | 19.9  | 23.5  | 26.1  | 30.2  | 31.1  | 28.8  | 24.1  | 18.9  | 13.5 |
|                                                                                  |             | MCWB   | 6.2   | 7.5   | 10.9  | 14.6  | 17.5  | 21.4  | 25.2  | 25.6  | 23.7  | 19.2  | 14.4  | 9.4  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 12.7  | 15.0  | 16.1  | 19.9  | 22.1  | 25.5  | 27.5  | 27.7  | 26.8  | 24.0  | 19.9  | 16.2 |
|                                                                                  |             | MCDB   | 13.5  | 16.3  | 17.0  | 20.8  | 24.1  | 29.0  | 31.5  | 32.0  | 29.8  | 25.5  | 21.0  | 17.4 |
|                                                                                  | 2%          | WB     | 9.8   | 11.7  | 14.7  | 18.2  | 20.9  | 24.5  | 26.7  | 27.1  | 26.1  | 22.5  | 17.8  | 13.2 |
|                                                                                  |             | MCDB   | 11.9  | 14.3  | 16.6  | 20.3  | 23.0  | 26.8  | 30.3  | 30.9  | 28.7  | 25.2  | 19.6  | 15.5 |
|                                                                                  | 5%          | WB     | 8.0   | 9.9   | 13.3  | 17.0  | 20.0  | 23.6  | 26.2  | 26.7  | 25.5  | 21.3  | 16.5  | 11.4 |
|                                                                                  |             | MCDB   | 10.5  | 12.3  | 15.8  | 19.5  | 22.5  | 25.8  | 29.4  | 30.2  | 28.2  | 24.2  | 18.9  | 13.8 |
|                                                                                  | 10%         | WB     | 6.8   | 8.3   | 11.9  | 15.9  | 19.1  | 22.7  | 25.8  | 26.2  | 24.9  | 20.2  | 15.4  | 10.0 |
|                                                                                  |             | MCDB   | 9.9   | 11.0  | 14.8  | 18.8  | 21.9  | 24.9  | 28.7  | 29.5  | 27.7  | 23.1  | 18.1  | 13.0 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.1   | 7.8   | 8.4   | 8.2   | 7.5   | 6.2   | 5.9   | 6.4   | 6.6   | 7.1   | 7.4   | 7.1  |
|                                                                                  |             | MCDBR  | 8.4   | 9.2   | 9.7   | 9.5   | 9.3   | 7.8   | 7.6   | 7.6   | 7.5   | 8.0   | 8.4   | 8.4  |
|                                                                                  |             | MCWBR  | 5.8   | 6.6   | 6.8   | 5.8   | 5.0   | 3.5   | 3.0   | 3.0   | 3.5   | 4.4   | 5.3   | 5.9  |
|                                                                                  | 5% WB       | MCDBR  | 7.4   | 8.1   | 8.3   | 7.3   | 7.2   | 5.9   | 6.5   | 6.6   | 6.4   | 6.9   | 7.1   | 7.3  |
|                                                                                  |             | MCWBR  | 6.4   | 7.1   | 7.3   | 5.7   | 4.9   | 3.2   | 2.8   | 2.9   | 3.4   | 4.6   | 5.8   | 6.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.375  | 0.420 | 0.479 | 0.525 | 0.535 | 0.557 | 0.548 | 0.528 | 0.486 | 0.460 | 0.409 | 0.375 |      |
|                                                                                  | taud        | 2.261  | 2.139 | 1.987 | 1.919 | 1.967 | 1.982 | 2.058 | 2.104 | 2.193 | 2.183 | 2.243 | 2.277 |      |
|                                                                                  | Ebn at Noon | 809    | 809   | 790   | 774   | 772   | 752   | 756   | 764   | 777   | 759   | 764   | 782   |      |
|                                                                                  | Edh at Noon | 106    | 134   | 170   | 190   | 184   | 181   | 166   | 156   | 137   | 127   | 108   | 99    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.71   | 3.34  | 4.25  | 4.90  | 5.30  | 4.70  | 5.22  | 5.36  | 4.17  | 3.28  | 2.67  | 2.41  |      |
|                                                                                  | RadStd      | 0.18   | 0.24  | 0.28  | 0.50  | 0.56  | 0.45  | 0.69  | 0.46  | 0.37  | 0.34  | 0.23  | 0.17  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +74     | N/A     |
| (47) | Regional (40 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page